



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Tuesday, 4 November 2025, 6:27 PM**State** Finished**Completed on** Tuesday, 4 November 2025, 6:28 PM**Time taken** 46 secs**Marks** 1.00/1.00**Grade** 4.00 out of 4.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int A[n];
8      for (int i = 0; i < n; i++) {
9          scanf("%d", &A[i]);
10     }
11
12     int k;
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16     while (i < n && j < n) {
17         int diff = A[j] - A[i];
18         if (diff == k && i != j) {
19             printf("1\n");
20             return 0;
21         } else if (diff < k) {
22             j++;
23         } else {
24             i++;
25         }
26     }
27
28     printf("0\n");
29     return 0;
30 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)